# 1.3 Subgroup, Quotient Group

June 12, 2026

**Definition 0.1.** Let $G$ be a topological group and $H \subseteq G$. Then $H$ is a subgroup of topological group $G$ if $H$ is a subgroup of group $G$ and $H$ is a closed set in topological space $G$.

*Remark* 0.1. $H$ is a topological group equipped with induced topology on $G$. A subgroup of an abstract group which is a topological group is a topological group. This works as restriction of continuous maps is continuous.

**Definition 0.2.** A normal subgroup $N$ of topological group $G$ if $N$ is a normal subgroup of abstract group $G$.

**Proposition 0.1.** Let $G$ be a topological group. Let $H$ be a subgroup of abstract group $G$.

(i) $\overline{H}$ is a subgroup of top. group $G$.

(ii) If $H$ is normal subgroup of $G$ then $\overline{H}$ is normal.

*Proof.* (i) Observe $\overline{H} = \bigcap_{V \in \mathcal{V}} HV$ where $\mathcal{V}$ is collection of all nbds of $e$. As $H \subseteq \overline{H}$ hence $\overline{H} \neq \emptyset$.

For $x, y \in \overline{H}$ we want to show $xy^{-1} \in \overline{H}$. We know $m : G \times G \to G$ is continuous, where $m(x, y) = xy^{-1}$. Observe $m^{-1}(\overline{H})$ is closed, and $H \times H \subseteq m^{-1}(\overline{H})$. Hence $\overline{H} \times \overline{H} = \overline{H \times H} \subseteq m^{-1}(\overline{H})$ and this implies $m(\overline{H} \times \overline{H}) \subseteq \overline{H}$. □

We construct a mapping

$$n \colon G \times G \to G, \quad (x, y) \mapsto xyx^{-1}$$

and observe that $n$ is a continuous map. This implies that $n^{-1}(\overline{H})$ is closed and similar to (Ci).

$$H \times H \subseteq n^{-1}(\overline{H}) \implies \overline{H} \times \overline{H} = \overline{H \times H}$$

is a subset of $n^{-1}(\overline{H})$. Hence

$$n(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

**Definition 0.3.** * Set of all left cosets set $G/H$. We have a natural map or projection

$$\pi \colon G \to G/H, \quad x \mapsto xH.$$

*Remark* 0.2. If $G$ is a topological group, we will topologize $G/H$ assuming $H$ is closed. Construct

$$\mathcal{T}_{G/H} = \{O \subseteq G/H \mid \pi^{-1}(O) \in \mathcal{T}_G\},$$

i.e., $O$ is open if and only if $\pi^{-1}(O)$ is open in $G$.

which ensures $\pi$ is a continuous map, moreover a quotient map where $\pi$ is surjective by construction.

**Lemma 0.1.** $G/H$ is a $T_1$-space, and $\pi$ is open.

*Proof.* Since $H$ is closed, then $\alpha H$ is closed by homeomorphism. Observe $G - \alpha H$ is open in $G$. Let $\tilde{\alpha} = \alpha H \in G/H$. Observe $\pi^{-1}(G/H - \tilde{\alpha}) = G - \alpha H$. Hence $\tilde{\alpha}$ is closed, $G/H$ is $T_1$.

We want to show $\pi$ is an open map. Let $O$ be an open map in $G/H$ then $\pi(O)$ is open in $G/H$ iff $\pi^{-1}(\pi(O))$ is open in $G$. Now $OH = \pi^{-1}(\pi(O))$ and $OH = \bigcup_{\alpha \in O} \alpha H$. Hence $OH$ is open. $\square$

$\pi^{-1}(\pi(O))$ is open which implies $\pi(O)$ is open.

**Lemma 0.2.** *$G/H$ is a $T_2$ space.*

*Proof.* Consider $xH \neq yH$ in $G/H$. Now $x \in xH$ and $x \notin G \setminus yH$ which is open as $H$ is closed and $yH$ is closed. Now, let $V$ be a neighborhood of $x$ such that $x \in V \subset G \setminus yH$. Observe $Vx \cap yH = \emptyset$ implies $VxH \cap yH = \emptyset$ otherwise $vh_1 = gh_2$ and $v = gh_2h_1^{-1}$ implying $Vx \cap yH \neq \emptyset$. Now, we can find a neighborhood $V_1$ of $V$ such that $V_1^{-1}V_1 \subset V$ therefore $V_1^{-1}V_1xH \cap yH = \emptyset$. This implies $V_1xH \cap V_1yH = \emptyset$, which is $\pi(V_1x) \cap \pi(V_1y) = \emptyset$ and by previous lemma $\pi(V_1x)$ and $\pi(V_1y)$ are open sets and $xH \in \pi(V_1x)$ and $yH \in \pi(V_1y)$. $\square$

*Remark* 0.3. We call $G/H$ the quotient space where $H$ is a closed subgroup of $H$, has equipped with quotient topology.

**Lemma 0.3.** *If $G$ is a topological group, $H$ is normal subgroup, the quotient topology equipped on the quotient group $G/H$ is a topological group.*

*Proof.* We want to show $\varphi_1 : G/H \times G/H \to G/H$ defined by

$$\varphi_1(\alpha_1, y_1) \mapsto \alpha_1 y_1^{-1}$$

is continuous.

And $\varphi : G \times G \to G$ defined by

$$\varphi(\alpha, y) \mapsto \alpha y^{-1}$$

is continuous. $\square$

$$\begin{array}{ccc} G \times G & \xrightarrow{\pi \times \pi} & G/H \times G/H \\ \downarrow \pi & & \downarrow \psi \\ G & \xrightarrow{\varphi} & G/H \end{array}$$

Observe that $\psi_1(\pi \times \pi) = \psi\pi$ and $\pi$, $\psi$ are continuous. Hence $\psi_1(\pi \times \pi)$ are continuous.

Let $O_1$ be an open subset of $G/H$ then $\psi_1(\pi \times \pi)^{-1}(O_1)$ is open. As $\pi \times \pi$ is an open map, observe that $\pi \times \pi(\pi \times \pi)^{-1}\psi_1^{-1}(O_1)$ is open. Hence $\psi_1^{-1}(O_1)$ is open.

*Proof.* $\square$ $\square$

**Lemma 0.4.** *Let $G$ be a topological group, and $H$ be a subgroup.*

(a) *If $G$ is compact then $H$ and $G/H$ are both compact.*

(b) *If $G$ is locally compact then $H$ and $G/H$ have both locally compact.*

*Proof.* (a) $H$ is a closed subset of a compact set and $G/H = \pi(G)$ where $G$ is compact and $\pi$ is a continuous mapping. Hence a continuous map of a compact set is compact.

(b) A closed subset of a locally compact set is locally compact.

Let $S$ be locally compact, $T = \overline{S}$ and $T \subseteq S$. For $p \in T \cap S$, there exists a neighborhood of $pinS$, $\overline{U_p}$ such that $\overline{U_p}$ is compact. Observe for $p \in T$, there exists an open set $T \cap U_p$ in $T$ such that $\overline{T \cap U_p} = \overline{T} \cap \overline{U_p}$ is compact as $\overline{T \cap U_p}$ is a closed subset of $\overline{U_p}$ which is compact. Hence $H$ is locally compact.

**Goal:** $G/H$ is locally compact.

□

Let $q_i \in G \setminus H$ and let $U_i$ be a neighborhood of $q_i$. Let

$$U = \pi^{-1}(U_i),$$

and let $x \in G$ such that $\pi(x) = q_i$. As $G$ is locally compact, we get a neighborhood $O$ of $x$ such that $\overline{O}$ is compact and $\overline{O} \subseteq U$ by the proposition given below. Then we have $\pi(\overline{O}) \subseteq \pi(U) = U_i$. If $O_i = \pi(O)$ is a neighborhood of $q_i$. And $\pi(\overline{O})$ is a continuous image of a compact set, hence we get $\pi(\overline{O})$ is compact. Since $G/H$ is $T_2$-space $\pi(\overline{O})$ is closed which implies

$$O \subseteq \overline{O} \Rightarrow O_i \subseteq \pi(\overline{O}) \quad \text{and} \quad \overline{O_i} \subseteq \pi(\overline{O}) = \pi(\overline{O}) \quad \text{which is compact, so is} \quad \overline{O_i}.$$

Thus $O_i$ is a neighborhood of $q_i$ such that $\overline{O_i}$ is compact. It follows $G/H$ is locally compact.

**Corollary 0.1.** If $G$ is locally compact, then $G/H$ is normal or $T_4$.

**Proposition 0.2.** For any topological group $G$, let $U$ be a neighborhood of $e$, there exists a neighborhood of $e$ $V$ such that $V \subseteq U$.

*Proof.* By properties (5) and (6) we know that for every neighborhood of $e$, $U$, we have a symmetric neighborhood of $e$ $V$ such that $V^2 \subseteq U$. Now $x \in \overline{V} = \bigcap_{w \in V} wV$ where $V^2$ is the set of all neighborhoods of $e$, this implies $x \in \overline{V} \subseteq V^2 \subseteq U$. □